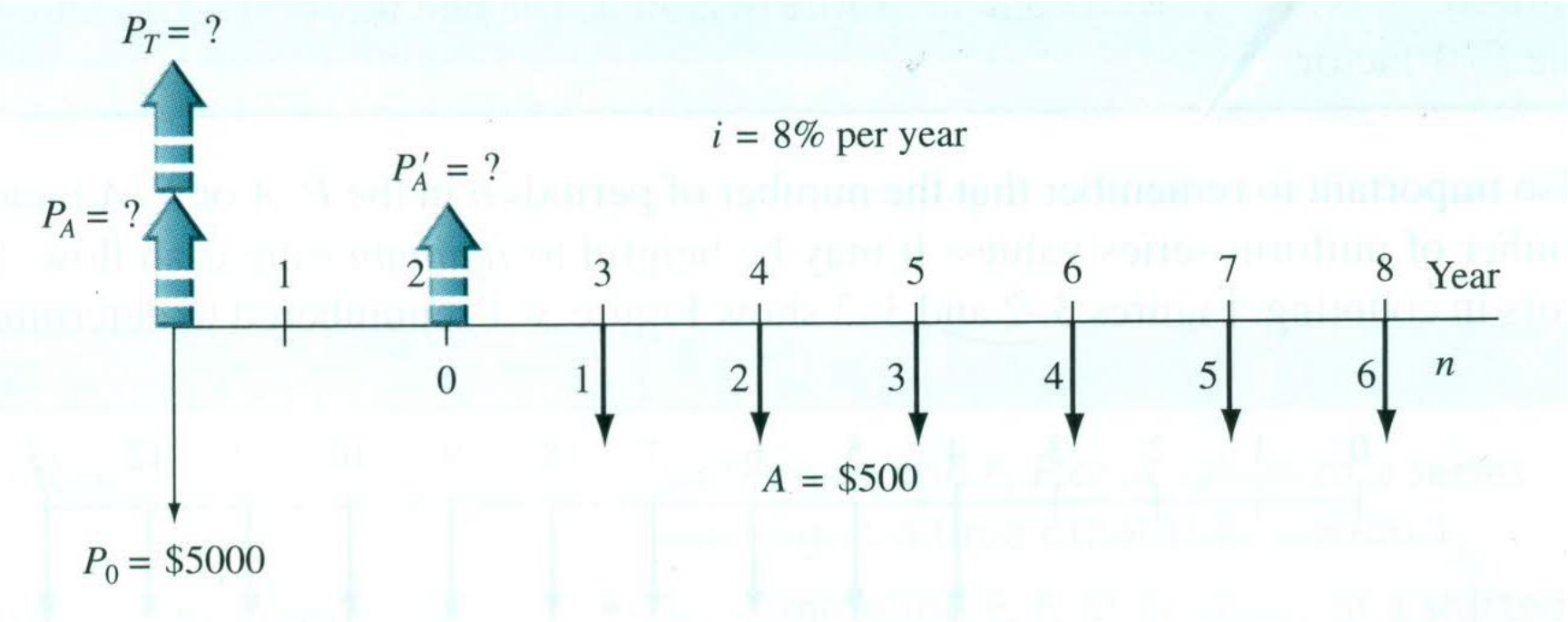
ME 491 Practice questions

ENGINEERING ECONOMY

Calculations for Uniform series that are shifted

- The offshore design group at Bechtel just purchased upgraded CAD software for $5000 now and annual payments of $500 per year for six years starting three years from now for annual upgrades. What is the present worth in year 0 of the payments if the interest rate is 8% per year?

CASH FLOW DIAGRAM



SOLUTION

- $P'_A = \$500(P / A,8\%,6)$
- Since P'_A is located in year 2, now find P_A in year 0.
- $P_A = P'_A\,(P/F,8\%,2)$
- The total present worth is determined by adding P_A and the initial payment *Po* in year O.
- $P_T = P_0 + P_A$
- $= 5000 + 500(P/A,8\%,6)(P/F,8\%,2)$
  $= 5000 + 500(4.6229)(0.8573)$
- $= \$6981.60$

QUESTION 2

- Recalibration of sensitive measuring devices costs $8000 per year. If the machine will be recalibrated for each of the six years starting 3 years after the purchase, calculate the 8-year equivalent uniform series at 16% per year.

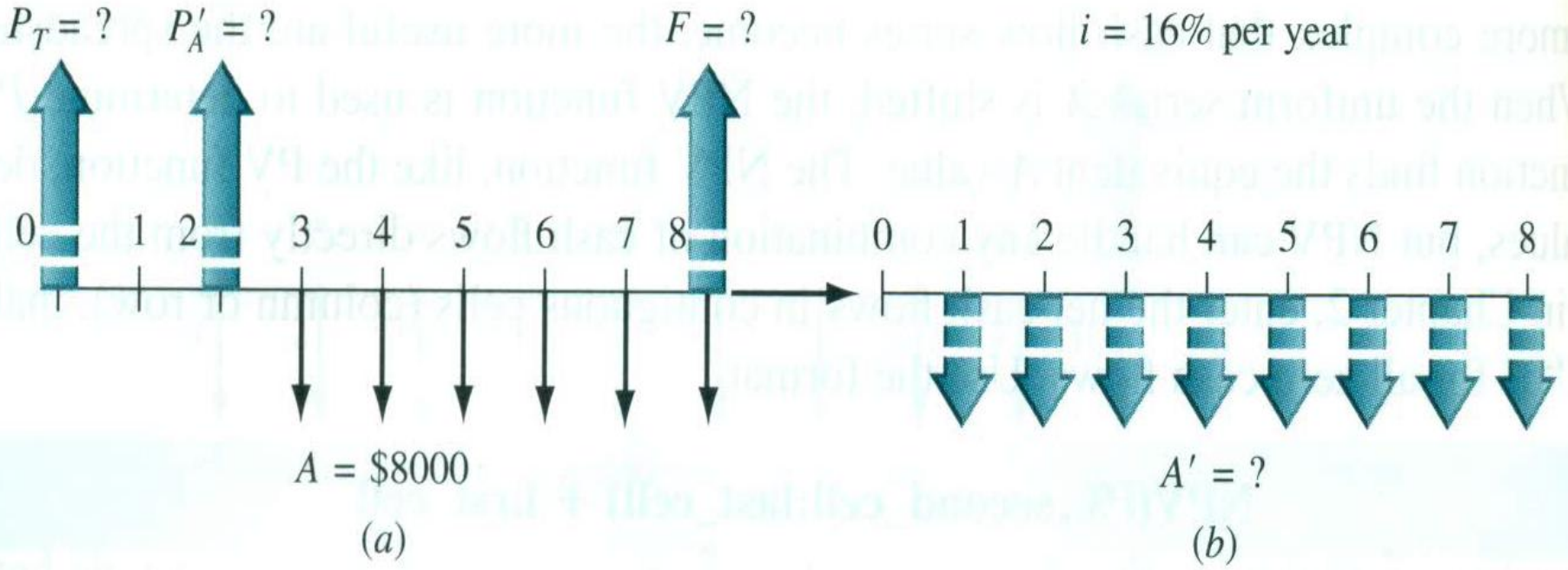
Present worth method

- $P'_A = 8000(P/A,16\%,6)$
- $P_T = P'_A(P/F,l6\%,2)$
  $=8000(P/A,16\%,6)(P/F,16\%,2)$
  $= 8000(3.6847)(0.7432) = \$21,907.75$
- The equivalent series *A' for* 8 *years* can now be determined via the *A/P* factor.
- $A' = P_r(A/P,16\%,8) = \$5043.60$

Future worth method

- First calculate the future worth *F* in year 8.
- *F = 8000(F/A16%,6)* = $71,820
- The *A/F* factor is now used to obtain *A'* over all 8 years.
- *A' = F(A/F,16%,8)* = $5043.20

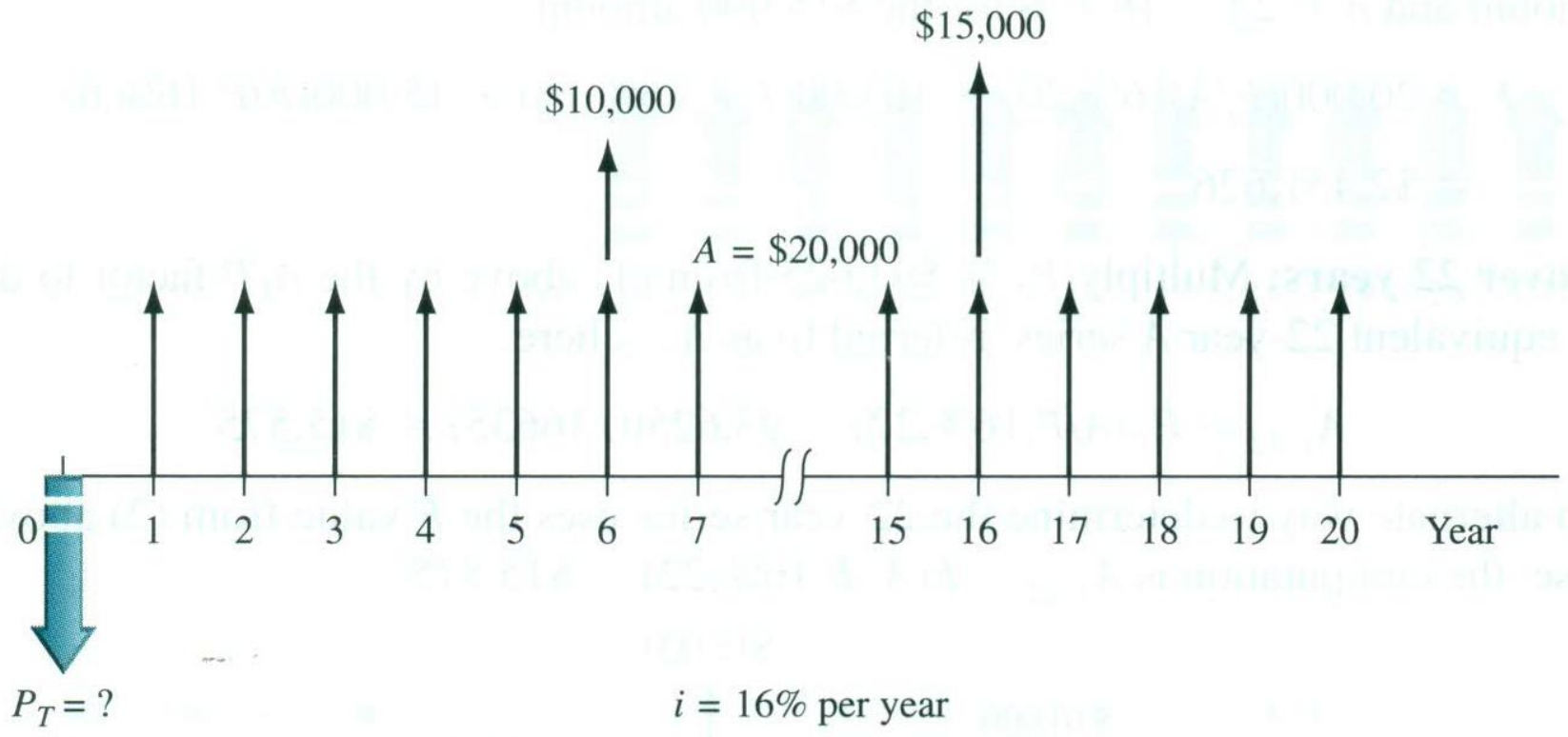
CASH FLOW DIAGRAM



Calculations involving uniform series and randomly placed single amounts.

- An engineering company in the Western Region that owns 50 hectares of valuable land has decided to lease the mineral rights to a mining company. The primary objective is to obtain long-term income to finance ongoing projects 6 and 16 years from the present time.
- The engineering company makes a proposal to the mining company that it pay $20,000 per year for 20 years beginning 1 year from now, plus $10,000 six years from now and $15,000 sixteen years from now. If the mining company wants to pay off its lease immediately, how much should it pay now if the investment is to make 16% per year?

CASH FLOW DIAGRAM



CASH FLOW COMPUTATIONS

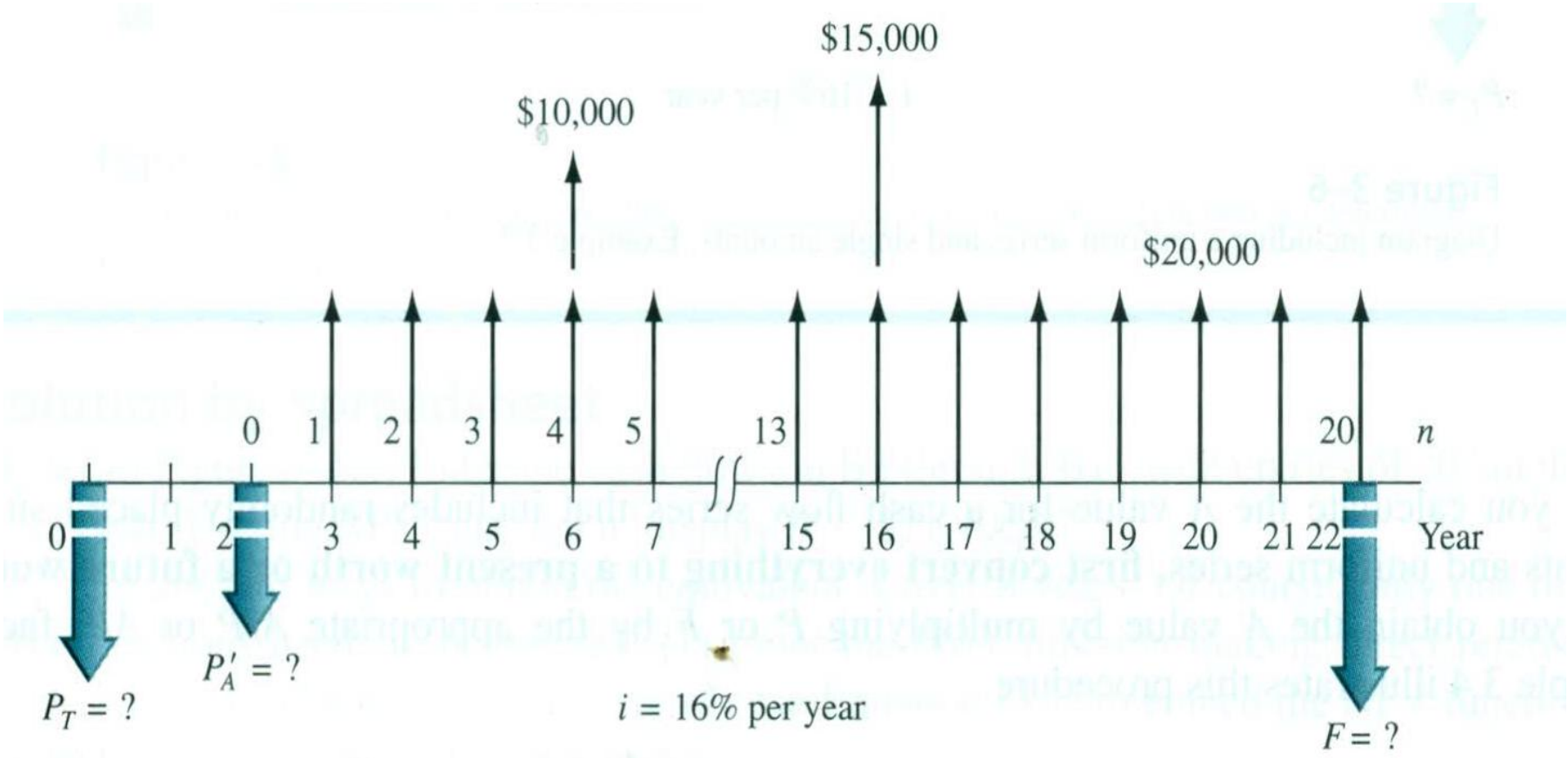
- $P_T = 20{,}000(P/A,16\%,20) + 1O{,}000(P/F,16\%,6) + 15{,}000(P/F,16\%,16)$

  = $124,075

MIXED FACTORS

- A design-build-operate engineering company in that owns a sizable amount of land plans to lease the drilling rights (oil and gas only) to a mining and exploration company.
- The contract calls for the mining company to pay $20,000 per year for 20 years beginning 3 years from now (i.e., beginning at the end of year 3 and continuing through year 22) plus $10,000 six years from now and $15,000 sixteen years from now.
- Utilize engineering economy relations to determine the five *equivalent values* listed below at 16% per year.
- Total present worth *P;* in year 0
- Future worth *F* in year 22
- Annual series over all 22 years
- Annual series over the first 10 years
- Annual series over the last 12 years

CASH FLOW DIAGRAM FOR THE DBOE COMPANY



Total present worth *P;* in year 0

- $P'_A = 20{,}000(P/A,16\%,20)$
- $P_T = P'_A(P/F,16\%,2) + 1O{,}000(P/F,16\%,6) + 15{,}000(P/F,16\%,16)$
  $= 20{,}000(P/A,16\%,20)(P/F,16\%,2) + 10{,}000(P/F,16\%,6)$
- $+ 15{,}000(P/F,16\%,16)$
- = $93,625

Future worth *F* in year 22

- *F* in year 22: To determine *F* in year 22 from the original cash flows, find *F* for the 20-year series and add the two *F* values for the two single amounts. Be sure to carefully determine the *n* values for the single amounts: *n* = 22 - 6 = 16 for the $10,000 amount and *n* = 22 - 16 = 6 for the $15,000 amount.
- *F = 20,000(F/A,16%,20) + 1O,000(F/P,16%,16) + 15,000(F/P,16%,6)*
- = $2,451,626